Multi-view Fusion Reverse Photography with Limited Sensing Resolution

Tawsif Ahmed, Bahaa Alattar, Gyungmin Ko, Victor Marcenac

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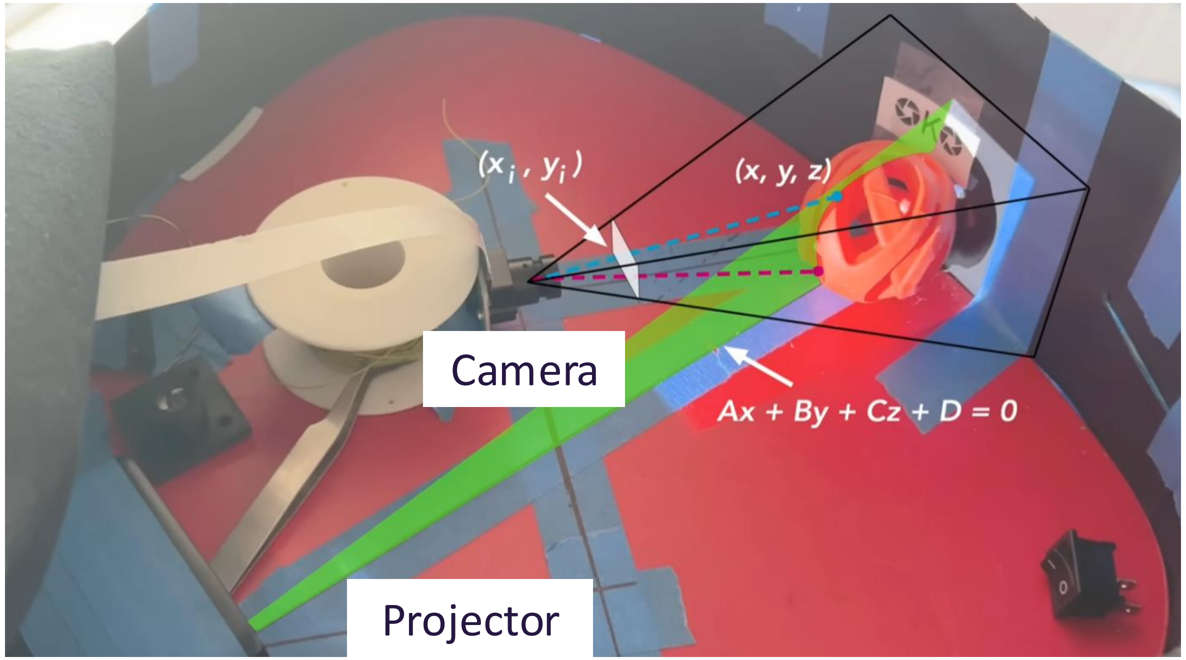
UNIVERSITY *of* WASHINGTON

Intro

Dual Photography is ...

> Measure the light transport between a projector and a camera with structured illumination, and you can re-render the scene from the projector's point of view



> Reference: Dual photography (Sen et al., 2005)

Our Approach

"Cleaning the 2D reconstruction with ML"

> **The reconstruction is noisy**
> - A hand-tuned median filter reduces some noise
> - But it cannot fully remove holes and speckle artifacts

> **Our ML approach**
> - An image-to-image network to denoise and inpaint the reconstruction
> - This is expected to generalize better than a fixed hand-tuned filter

Raw reconstruction

After median filter

Hardware

> 1 Projector (1280x700)
> 2 Intel RealSense Cameras (D455)
> Black Foam Boards

*for its incredible 1 MP RGB Camera only

Visit the Intel Store

Intel RealSense D455 Webcam - 90 fps - USB 3.1-1280 x 800 Video

4.1 (25) | Search this page

$459.00

Or $114.75 /2 weeks (x4). Select from 2 plans

Get a $150 Gift Card: Apply and pay ~~$459.00~~ **$309.00** if approved for Prime Visa

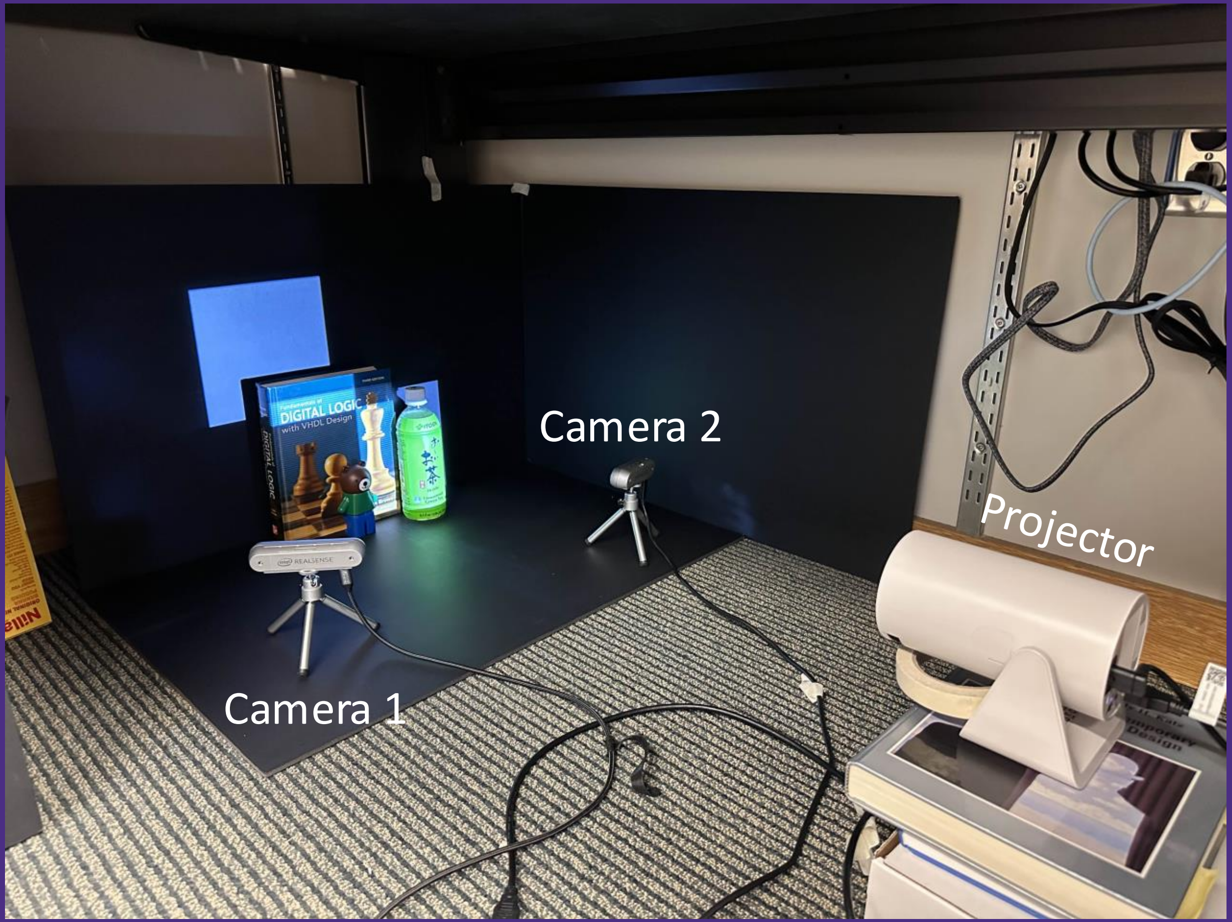
FREE Returns

Brand	Intel
Photo Sensor Technology	CCD
Video Capture Resolution	1280 x 800
Maximum Focal	100 Millimeters

Hardware Setup

- Two cameras facing object with the projector projecting behind it
- Closed-off environment to prevent external light source

Closed-off environment



General Hardware Setup

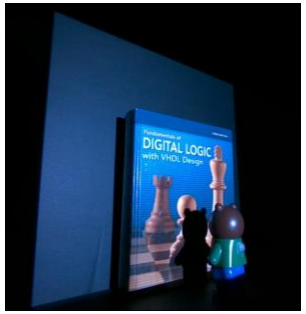
Algorithm

Projector View Synthesis

- **Split projector into 256 × 256 cells**
  - **Give each cell a unique 16-bit code**
- **Project code bits as 16 binary patterns**
  - **Capture each pattern and its complement**
  - **Subtract complements to isolate signal**
- **Decode each camera pixel's 16-bit code**
  - **Map camera pixels back to projector cells**

Reference used to get illuminated color

00	01
11	10

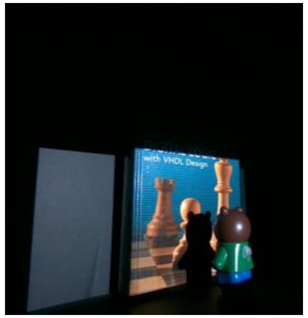


Pattern 1

00	01
11	10

Pattern 2

00	01
11	10



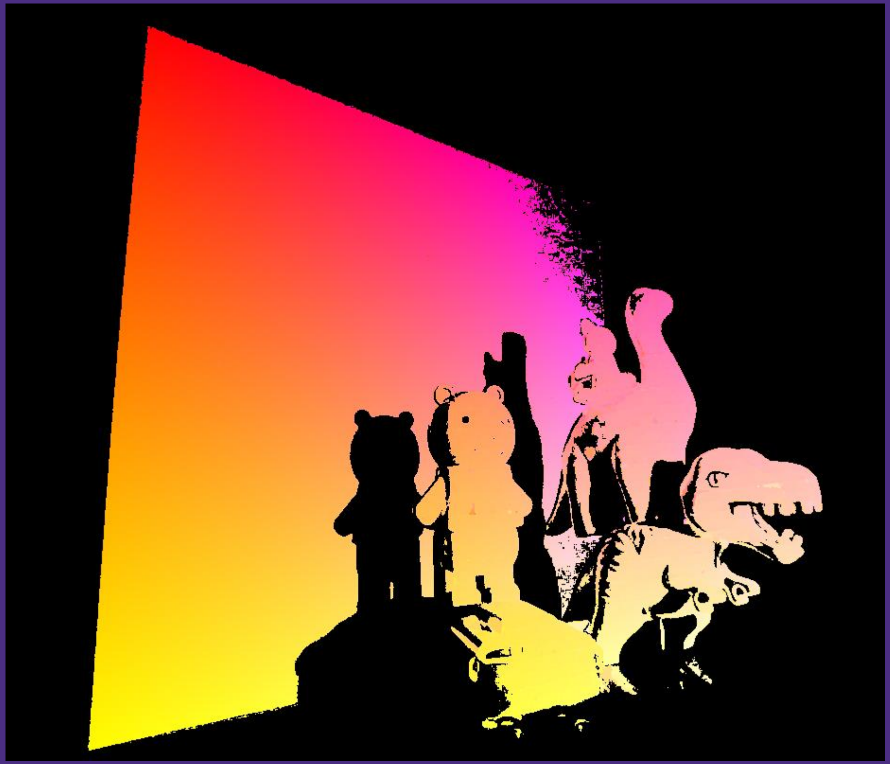
Algorithm

Reconstruction

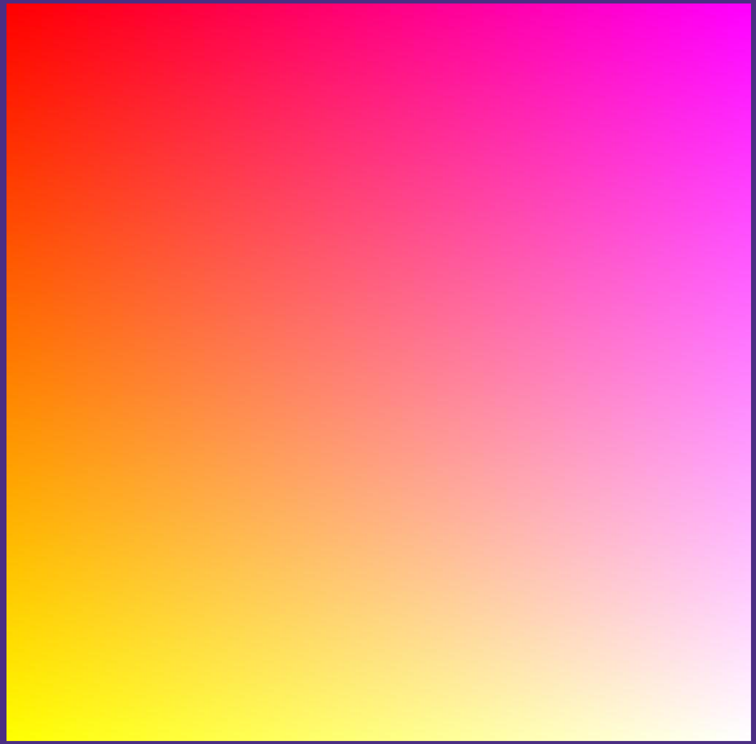
- Decode & validate each pixel's 16-bit code
  - Threshold to bits and drop low-contrast/dark pixels
- Clean up the code map & group pixels
  - Median filter removes noise
  - Average each cell's pixels
- Render & normalize projectors image
  - Map average values onto the projector grid

Perspectives of what the projector sees…

Left Camera

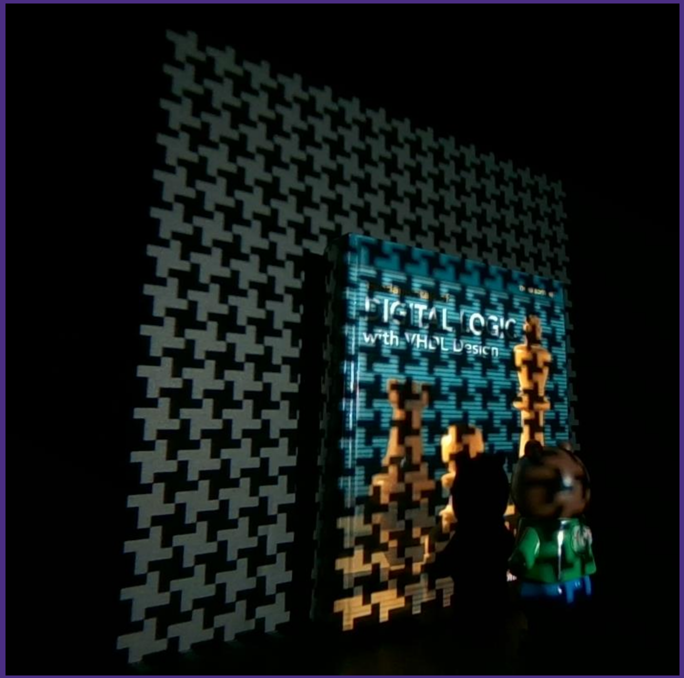


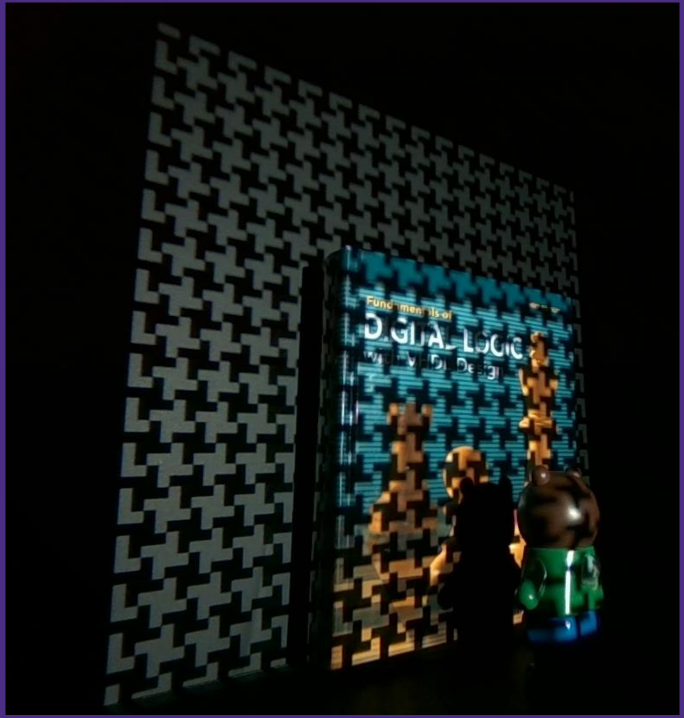
Projector



Right Camera

Original Pattern

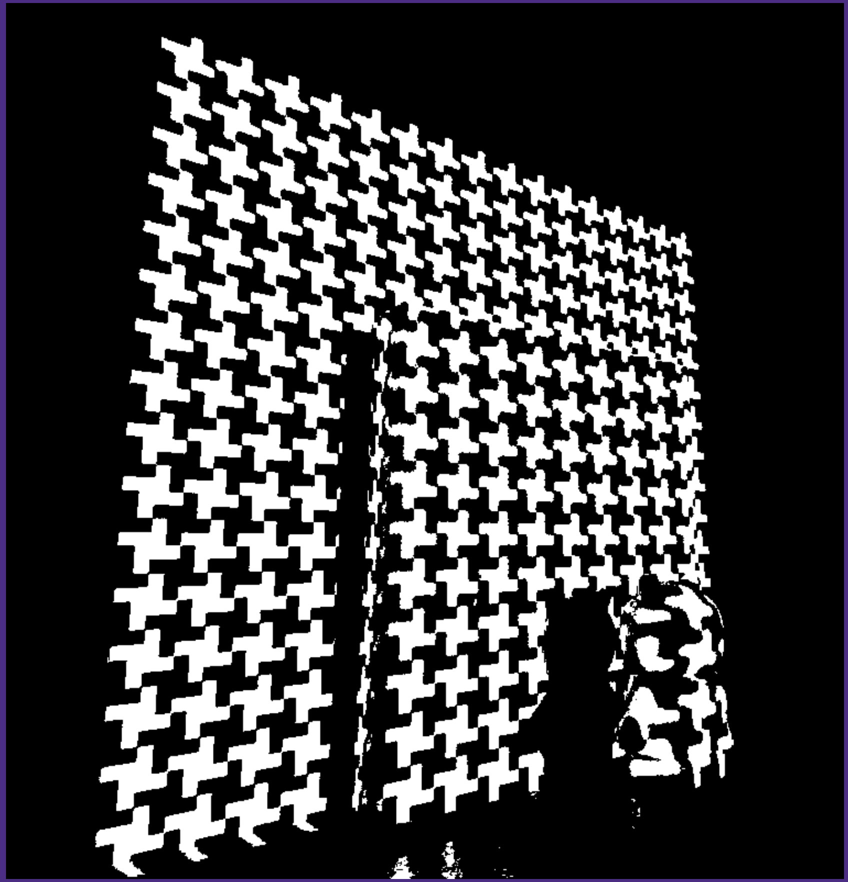




Complement Pattern

Confidence Map

Combined Bitmask
for this layer



Single Camera

No Complement

W/ Complement

*W/ Complement

Fusion Synthesis

Single Camera

Dual Camera

Total Improvement

Single Camera/Pattern

Dual W/ Complement

Limitations

> Projector Resolution: 1280x700
> Camera Resolution: 1280x800

> Projection occupies ~1/3 of camera
> Projection grid: 256x256 cells

> 1280x800 / 3 / (256x256) = **5 pxls / cells**
> Some cells are smaller

VERY SENSITIVE TO NOISE -->

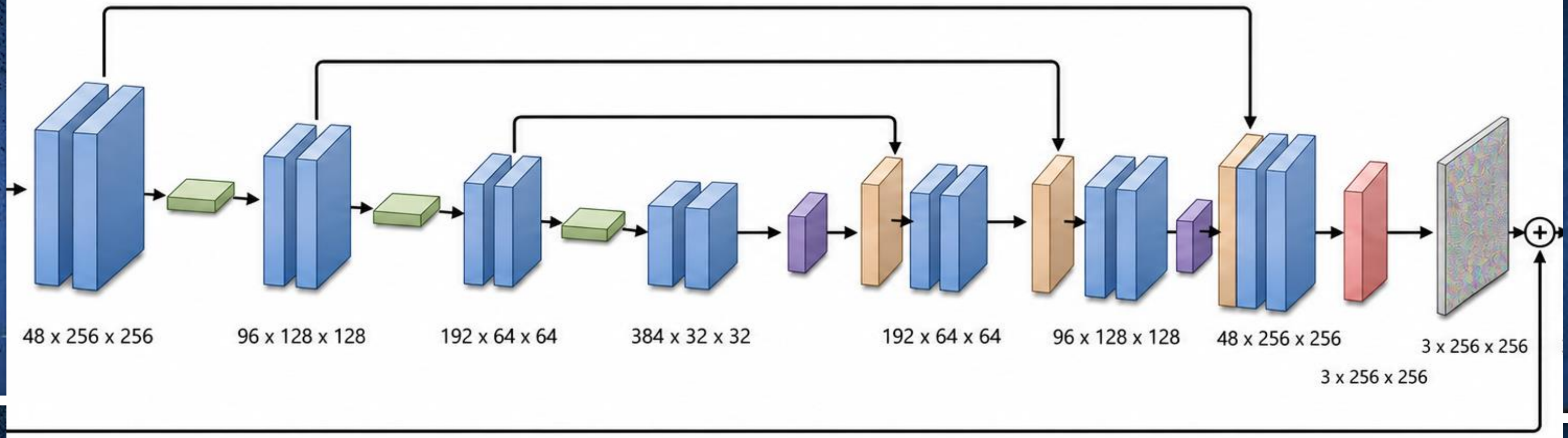
Extension: CNN Denoising

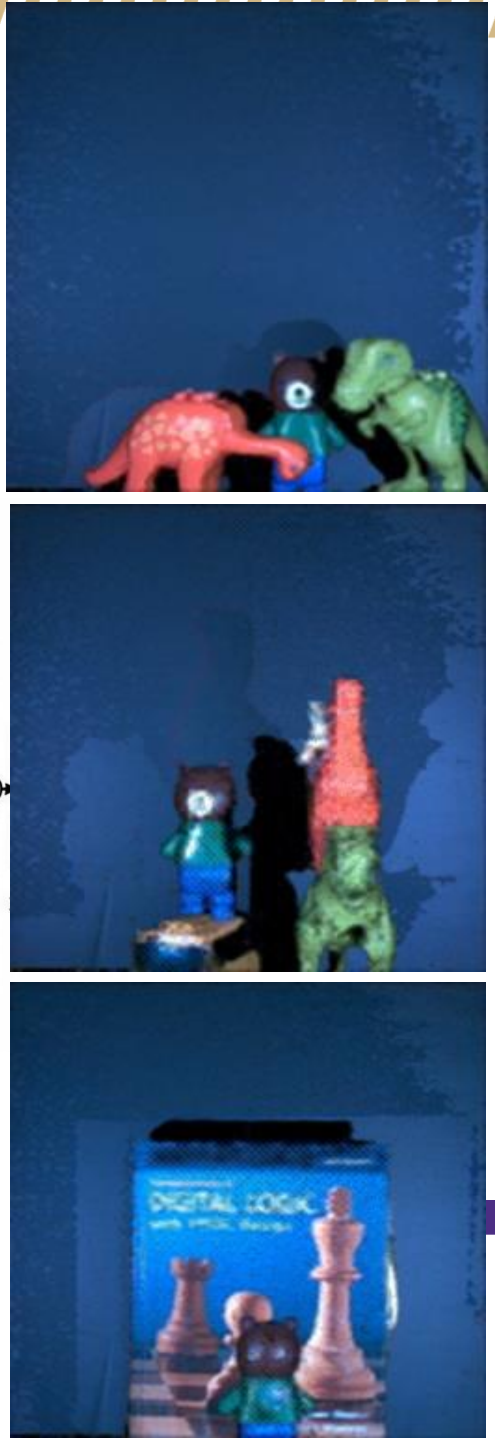
Training Dataset:

- Up-sample COIL - 100: 128x128 --> 256x256
- Synthetic noise:
  - Section image into clusters of 64x64, ..., 4x4
  - Permute pixels with increasing probability for smaller clusters
    - Global: 10% Swap pixels, 3% Kill to Black
    - Top half has higher noise concentration

Before After

ML–Based Denoising results





Reference

- **Dual Photography: https://dl.acm.org/doi/10.1145/1186822.1073257**
- **Single-Pixel Imaging Survey: https://pmc.ncbi.nlm.nih.gov/articles/PMC5557330/**
- **OkiOptics Demo (Single Pixel): https://youtu.be/dMH6VUs5u8k**
- **OkiOptics Demo (Multi-View): https://youtu.be/TcXMf0mTh94**